

KOTLIN MULTIPLATFORM

Collaborative Todo

One shared Kotlin codebase. Real-time, shareable to-do lists running natively on Android, iOS & Desktop — and on the Web.

Android

iOS

Desktop (JVM)

Web

Compose Multiplatform

Ktor server

One codebase, every platform

Every screen in this document was rendered from the **same shared Compose UI** that ships to each device — not a mockup.

The clients share everything below the view layer — domain model, networking, and ViewModels — so behaviour is identical wherever the app runs (ADR-0001).

Android & iOS & Desktop

Single shared Compose Multiplatform UI. iOS can drop to SwiftUI per-screen when fidelity demands it.

Web

Compose HTML (real DOM, selectable text, accessible) — reusing the exact same shared ViewModels.

:common

DTOs, routes & domain types shared by client and server.

:client-core

ViewModels, API client & state — the logic every client runs.

:ui-compose

Shared Compose screens for Android / iOS / Desktop.

:server

Ktor + Postgres backend with real-time change stream.

How these images were produced: each screen is captured by a render test that feeds canned data into the real composables and rasterises the Compose tree offscreen — so the demo can never drift from the shipping UI.

Passwordless sign-in

Enter an email, receive a one-time code, you're in. No passwords to manage.

Collaborative Todo

Sign in with your email

Send code

Collaborative Todo

Enter the code sent to felix@theplant.jp
Dev code: 123456

Verify

Use a different email

Step 1 — request a code by email

Step 2 — verify the 6-digit code

Your lists & joining shared ones Slice 5

Create your own lists, or paste an invite link/code to join a list someone shared with you.

Lists

3

New list

Add

Trip to Kyoto

Team sprint

Groceries

Invite link or code

Join

Sign out

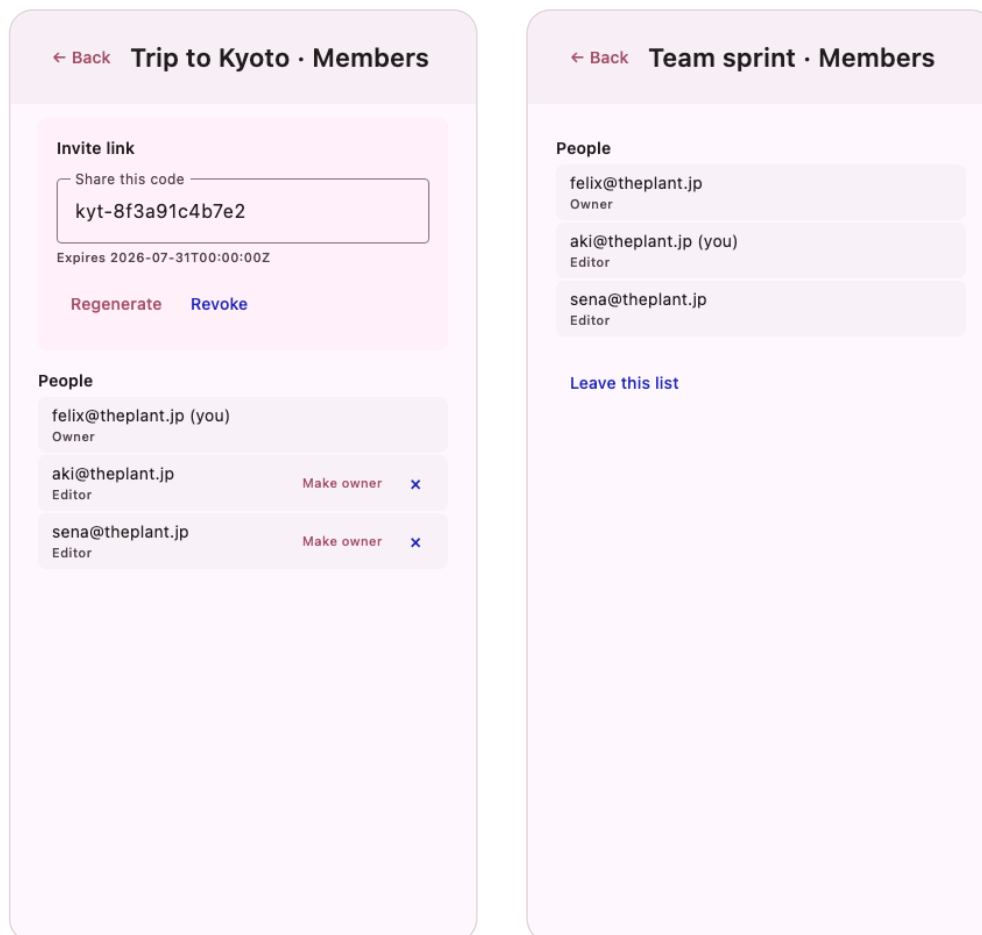
Delete account

Lists index — create, rename, delete, or join by invite

Sharing via invite link Slice 5

The owner generates a single, revocable invite link. Anyone who follows it joins the list as an editor.

- **Owner-only controls** — generate, regenerate, or revoke the link; optional expiry.
- **Roster at a glance** — see everyone on the list and their role.
- **Unguessable tokens** — the link embeds an opaque token, never a list id.

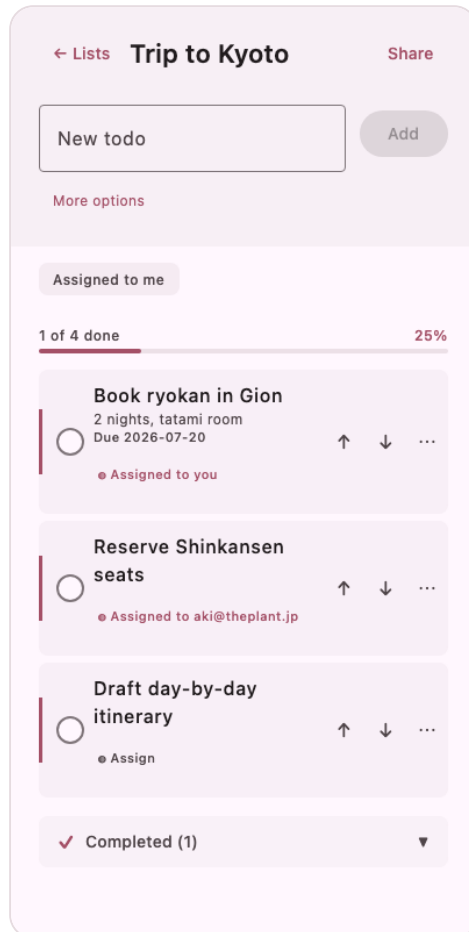


Owner view — active invite link + member roster Editor view — sees the roster, can leave the list

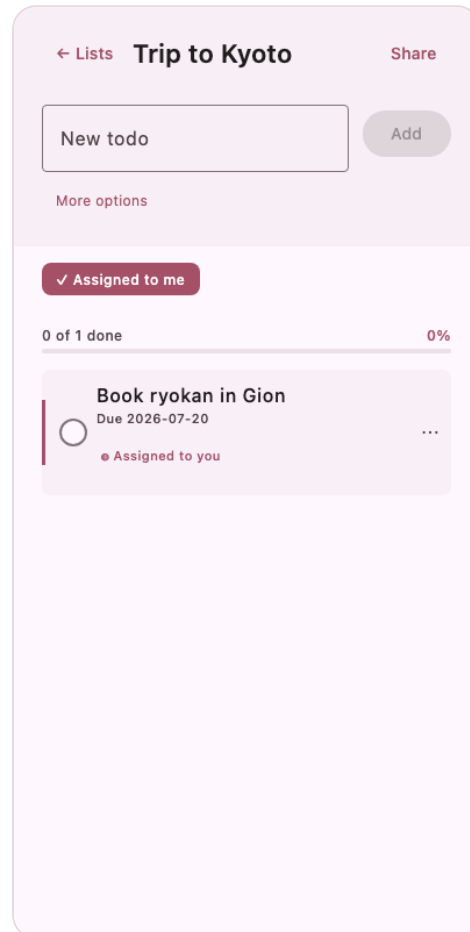
Assigning todos to teammates Slice 6

Assign any todo to a member, and filter the list down to just what's on your plate.

- **Assignee picker** — tap a todo's assignee chip to hand it to any member (or unassign).
- **“Assigned to me”** — one tap filters the whole list to your work.
- **Progress at the top** — done/total and a completion bar update as work lands.



A shared list with per-todo assignees



Filtered to “Assigned to me”

Real-time updates Slice 8

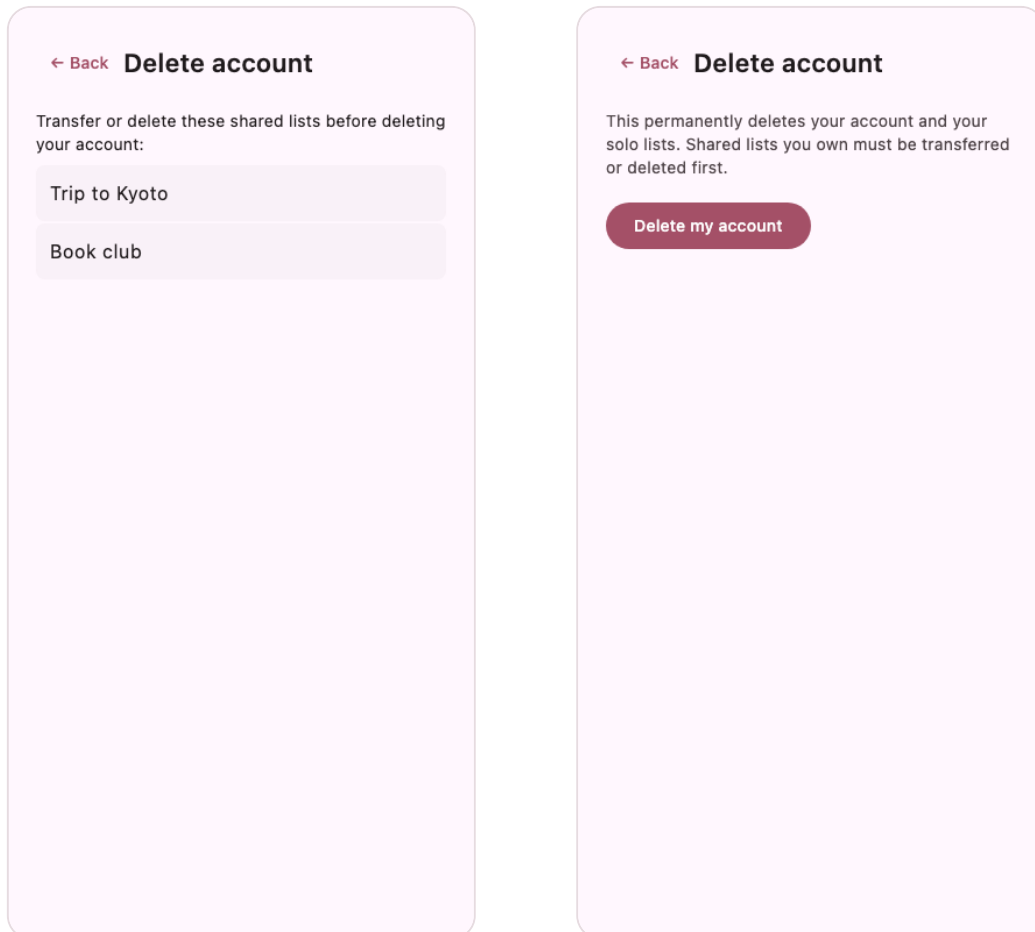
Changes made by one member appear on every other member's device within moments — no refresh.

The server pushes a change signal over Server-Sent Events (ADR-0006); each client re-fetches the affected list automatically and reconnects if the stream drops. The transport is isolated behind a single client seam, swappable for WebSockets if needed. The same live behaviour runs on all four platforms because it lives in the shared ViewModels.

Safe membership lifecycle Slice 7

A shared list always has exactly one owner — so leaving, transferring, and account deletion are guarded by invariants (ADR-0009).

- **Transfer ownership** — an owner can hand a list to another member and become an editor.
- **Deletion is blocked** when you still own shared lists — you must transfer or delete them first.

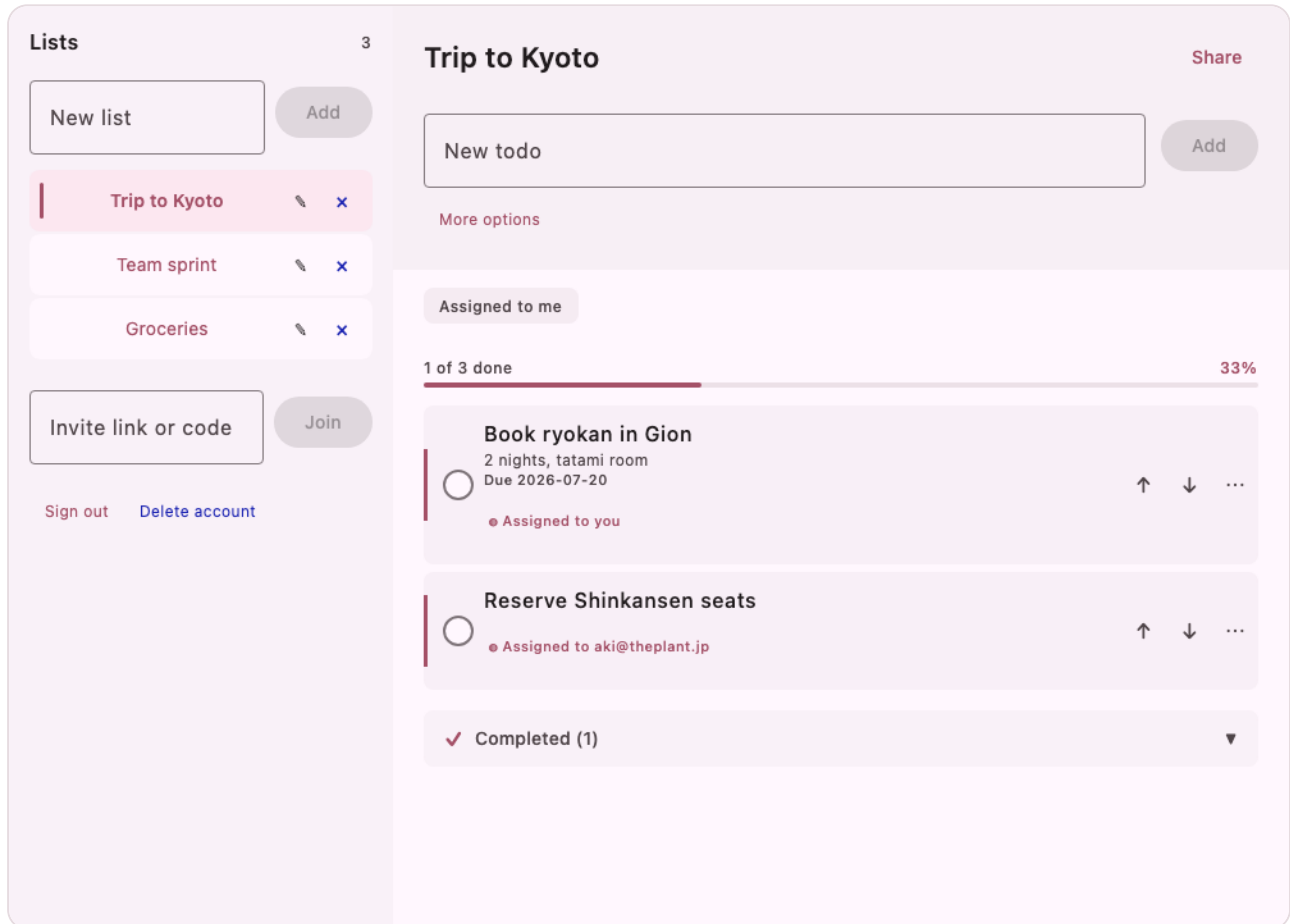


Deletion blocked — shared lists must be handled first

Nothing shared — safe to delete

Same app, desktop form factor

On a wide screen the shared UI lays out as a master-detail workspace — the identical composables that render the phone screens, given more room.



Desktop (JVM) — lists sidebar + live todo detail, one shared UI